

Density is not a static property but can increase or decrease over time as individuals are added to or removed from a population. As seen on the opening page of this chapter, additions occur through birth (which we define here to include all forms of reproduction) and **immigration**, the influx of new individuals from other areas. The factors that remove individuals from a population are death (mortality) and **emigration**, the movement of individuals out of a population and into other locations.

While birth and death rates influence the density of all populations, immigration and emigration can also have substantial effects. For example, studies of a population of Hector's dolphins (see Figure 53.2) in New Zealand showed that immigration accounted for about 15% of the total population size each year. Emigration was also important, typically occurring during the winter season when the dolphins move farther from shore. In general, both immigration and emigration represent key factors affecting the populations of many species. Emigration, for example, not only reduces the density of the original population, it also can result in the establishment of new populations in regions of favorable habitat, as we'll discuss later in this chapter.

➔ **Mastering Biology Animation: Techniques for Estimating Population Density and Size**

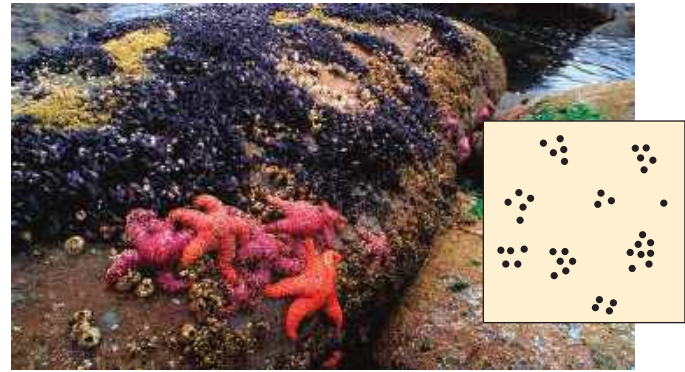
Patterns of Dispersion

Within the boundaries of a population, the spacing among individuals may differ substantially, creating contrasting patterns of dispersion. Such differences in spacing can provide insight into the biotic and abiotic factors that affect individuals in the population.

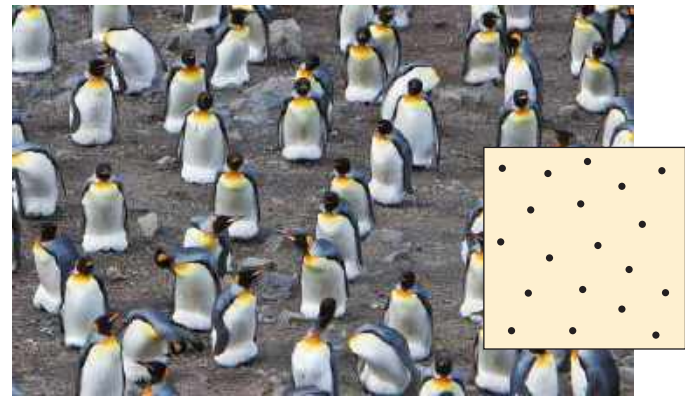
The most common pattern of dispersion is *clumped*, in which individuals are aggregated in patches. Plants and fungi are often clumped where soil conditions and other environmental factors favor germination and growth. Mushrooms, for instance, may be clumped within and on top of a rotting log. Insects and salamanders may be clumped under the same log because of the higher humidity there. Clumping of animals may also be associated with mating behavior. Sea stars group together in tide pools, where food is readily available and where they can breed successfully (Figure 53.3a). The aggregation of individuals into groups may also increase the effectiveness of predation or defense; for example, a wolf pack is more likely than a single wolf to subdue a moose, and a flock of birds is more likely than a single bird to warn of a potential attack.

A *uniform*, or evenly spaced, pattern of dispersion may result from direct interactions between individuals in the population. Some plants secrete chemicals that inhibit the germination and growth of nearby individuals that could compete for resources. Animals often exhibit uniform dispersion as a result of antagonistic social

▼ **Figure 53.3** Patterns of dispersion within a population's geographic range.



(a) **Clumped.** Sea stars group together where food is abundant.



(b) **Uniform.** King penguins and other birds that nest on small islands often exhibit uniform spacing, maintained by aggressive interactions between neighbors.



(c) **Random.** Dandelions grow from windblown seeds that land at random and later germinate.

❓ Patterns of dispersion can depend on scale. How might the penguin dispersion look from an airplane flying high above the ocean?

interactions, such as **territoriality**—the defense of a bounded physical space against encroachment by other individuals (Figure 53.3b).

In *random* dispersion (unpredictable spacing), the position of each individual in a population is independent of other individuals. This pattern occurs in the absence of strong